

ARI 2129 - Principles of Computer Vision for AI

Adversarial Quiz: Rationale Document

Topic: Point Processing - Histogram Equalisation and CLAHE

Link to Google Form:

https://docs.google.com/forms/d/e/1FAIpQLSewYt7Hf_c2F_Puk_2qmEQdqsVdyPw9yZXImYgbKWmNDWSEEQ/viewform?usp=header

Q1. What does histogram equalisation actually do to an image?

Correct answer (A): It remaps existing pixel intensities using the image's cumulative distribution, so values spread more widely across the available range. HE re-labels intensity values without inventing information and without looking at where pixels sit.

Option B. Confuses contrast with spatial resolution; HE never changes pixel positions or image dimensions.

Option C. The most tempting wrong answer. If detail was lost when the photo was taken, HE cannot bring it back. It can only stretch what is already there.

Option D. Averaging neighbouring pixels is denoising, not equalisation; HE uses no spatial neighbours.

Q2. Why use the CDF as the transformation function, rather than the histogram itself?

Correct answer (A): The CDF is monotonically non-decreasing, so it preserves the ordering of intensities. If a brighter input could become darker after equalisation, the output would not be a valid image; only a non-decreasing function rules that out, and the CDF always is.

Option B. The CDF has the same number of entries as the histogram (256 for 8-bit), so nothing is being compressed.

Option C. The widespread "always flat" myth; the output of HE is only approximately uniform because intensities are discrete integers.

Option D. The CDF cannot be faster than the histogram, because computing it requires the histogram first plus a cumulative sum.

Q3. For an 8-bit image ($L = 256$), which formula is correct?

Correct answer (D): $s_k = \text{round}(255 \times \text{CDF}(r_k))$. The general formula is $s_k = \text{round}((L - 1) \times \text{CDF}(r_k))$; for 8-bit images $L - 1 = 255$, because intensities run from 0 to 255 inclusive.

Option A. The classic off-by-one. With $\text{CDF} = 1$ this gives 256, which is not a legal pixel value.

Option B. Swaps $p(r_k)$ (probability at one level) for $\text{CDF}(r_k)$ (cumulative). The resulting mapping is not monotonic.

Option C. Reverses the scaling direction; CDF is already in $[0, 1]$, so dividing by 255 gives values near zero.

Q4. Compute the equalised value of input level 2 in a 4-level image with counts (2, 3, 4, 1) out of 10 pixels.

Correct answer (C): 3. Probabilities $p = (0.2, 0.3, 0.4, 0.1)$; the CDF is the running sum (0.2, 0.5, 0.9, 1.0); so $\text{round}((4 - 1) \times 0.9) = \text{round}(2.7) = 3$.

Option A. Uses $p(2) = 0.4$ instead of $\text{CDF}(2) = 0.9$: $\text{round}(3 \times 0.4) = 1$. The PDF-vs-CDF slip again.

Option B. Forgets the $(L - 1)$ scaling and just reads the input back, or rounds the CDF directly.

Option D. Uses $L = 4$ instead of $(L - 1) = 3$: $\text{round}(4 \times 0.9) = 4$, an illegal output in a 4-level system (max valid = 3).

Q5. After HE on an 8-bit photo, why is the output histogram not perfectly flat?

Correct answer (D): Because intensities are discrete integers, multiple input levels can be sent to the same output level, leaving the output only approximately uniform. The flatness result from probability theory applies in the continuous case; rounding to 256 levels collapses inputs and leaves gaps.

Option A. Calling it a bug is exactly the misconception this question targets; the theory only promises uniformity in the continuous domain.

Option B. "HE flattens the CDF, not the histogram" sounds clever but HE uses the CDF as the mapping and does change the output histogram.

Option C. Invented; HE behaves the same on a single channel regardless of how many channels the image has.

Q6. What problem is solved by the corrected formula that subtracts CDF_min?

Correct answer (A): It forces the darkest intensity actually present in the image to map to 0, preventing a washed-out look when the lowest bins are empty. If the darkest pixel is at level 50, the simple formula would map level 50 to some non-zero value and the whole image would come out brighter than it should.

Option B. Irrelevant; this correction is a per-channel brightness fix, not anything to do with combining RGB channels.

Option C. Tempting to assume any correction is about noise, but noise amplification is what CLAHE's clip limit handles.

Option D. Flatness is a separate property; the correction only shifts where the output range begins.

Q7. What is the clip limit in CLAHE for?

Correct answer (B): It restricts how aggressively contrast can be amplified in nearly uniform regions, which would otherwise turn sensor noise into visible texture. A very tall histogram bin in a tile usually means a smooth area; without clipping, the local CDF rises steeply there and stretches tiny noise differences into visible contrast.

Option A. Block artifacts between tiles are handled by bilinear interpolation across tile centres, not by clipping.

Option C. Clipping does not reduce the number of histogram bins; an 8-bit tile still has 256 of them.

Option D. Backwards; clipping deliberately holds back contrast rather than encouraging full-range output.

Q8. How does CLAHE prevent block artifacts at tile boundaries?

Correct answer (D): By blending the mappings of the four nearest tile centres using bilinear interpolation, weighted by distance to each centre. A pixel near a tile centre is dominated by that tile's mapping; a pixel between tiles gets a smooth blend. Edge pixels use fewer neighbours.

Option A. Averaging a pixel with eight neighbours is spatial smoothing; CLAHE never averages pixel intensities, only mapping functions.

Option B. The clip limit is one number per tile, not something that varies pixel-by-pixel; this option invents a mechanism.

Option C. Using a global histogram would remove the adaptiveness that makes CLAHE useful, defeating the point of the method.

Q9. CLAHE with a 16×16 grid and high clip limit gives a grainy low-light image. What should the team change?

Correct answer (D): Lower the clip limit and consider a larger tile region (say 8×8) so each local histogram has more data and is more stable. The grainy look in smooth regions is textbook noise amplification, and the clip limit is the parameter that controls it directly.

Option A. Raising the clip limit pushes CLAHE toward ordinary AHE, exactly where noise gets worse.

Option B. Smaller tiles plus a high clip limit is the worst possible combination for noise: less stable histograms, steeper CDFs.

Option C. Interpolation is not the source of grain; it is the fix for tile boundaries. Turning it off replaces one problem with a worse one.

Q10. Why might CLAHE help U-Net segmentation but not Mask R-CNN on the same X-rays (Ahmad et al., 2025)?

Correct answer (B): A nicer-looking input is not always a better input for a downstream model. A stronger architecture may already be robust to contrast variation, so preprocessing gains can be small or absent. U-Net seems to have benefited from sharper local contrast; Mask R-CNN had likely already learned features that did not need it.

Option A. "Fundamentally incompatible" over-generalises from one study; the two are not architecturally in conflict.

Option C. Calling the result a bug is the very assumption the study disproves: enhancement does not have to help every model.

Option D. Both U-Net and Mask R-CNN were applied to the same images, so the grayscale-vs-RGB explanation has no foothold.